# Distribution

**Question no 1:-** Simulate 30 rolls with =RANDBETWEEN(1,6). What is the probability of rolling a 3 exactly 5 times? (Hint: Use BINOM.DIST)

Answer:- **Identify parameters**

- Number of trials (n) = 30 rolls
- Probability of success (p) = Probability of rolling a 3 = 16
- Number of successes (k) = 5

## Formula (Binomial Distribution)

$P(X=k)=(n/k)\cdot p^k\cdot(1-p)^{n-k}$

## Substitute values

$P(X=5)=(30/5)\cdot(1/6)^5\cdot(5/6)^{25}$

## Excel function

In Excel, you can directly compute this using:

Code

=BINOM.DIST(5, 30, 1/6, FALSE)

- `5` → number of successes
- `30` → number of trials
- `1/6` → probability of success
- `FALSE` → exact probability (not cumulative)

The probability of rolling a **3 exactly 5 times in 30 rolls** is given by:

$P(X=5)=(30/5)\cdot(1/6)^5\cdot(5/6)^{25}$

**≈ 0.186 (18.6%)**

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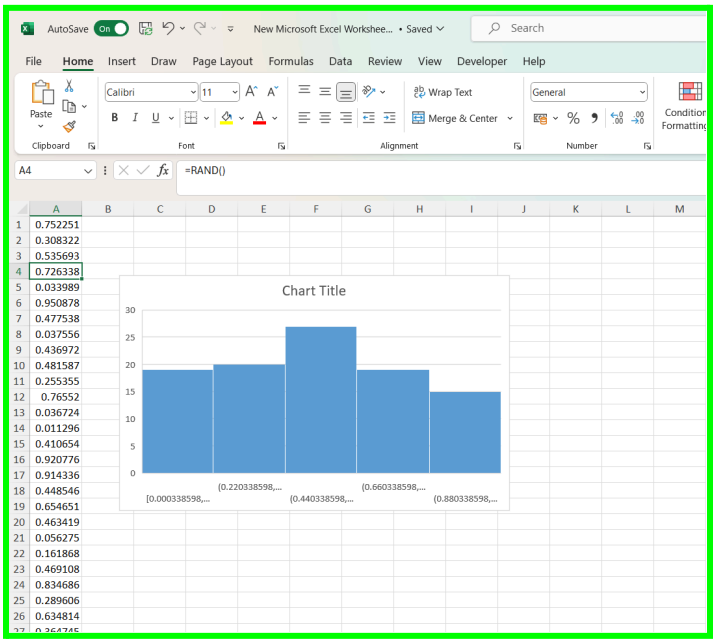
**Question no 2:-** Generate 100 values in Excel using the continuous uniform distribution RAND() and plot a histogram. Describe the shape of the distribution.

**Answer:-** We generate 100 values in Excel using the continuous uniform distribution with `=RAND()`, each value will lie between 0 and 1.

## Plotting the histogram

# Distribution

- Insert the 100 values into a column.
- Use Insert → Charts → Histogram to plot the frequency distribution.




## Shape of the distribution

- Since `RAND()` produces values uniformly between 0 and 1, the histogram will show a roughly flat shape.
- Each interval (bin) should have approximately the same frequency, though small fluctuations will appear because of randomness.
- As the sample size increases (say 1,000 or 10,000 values), the histogram will look more and more like a perfect rectangle.

The histogram of 100 values generated by `RAND()` will be approximately uniform — meaning all ranges between 0 and 1 are equally likely. With only 100 values, you'll see some unevenness, but overall it will resemble a flat distribution.

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**Question no 3:-** A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?

**Answer:-** This is a direct application of the Empirical Rule (68–95–99.7 rule) for normal distributions.

### Information given

- Mean (μ) = 50
- Standard deviation (σ) = 5
- Interval: 45 to 55

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## Relationship to standard deviation

$45 = \mu - \sigma$ and $55 = \mu + \sigma$

So the interval is within one standard deviation of the mean.

## Empirical Rule

For a normal distribution:

- About 68% of values lie within $\mu \pm 1\sigma$.
- About 95% lie within $\mu \pm 2\sigma$.
- About 99.7% lie within $\mu \pm 3\sigma$.

Approximately 68% of the values lie between 45 and 55.

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**Question no 4:-** What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset.

**Answer:-** **Concept of Standardization (Z-score)**

**Standardization** (or computing a **z-score**) is the process of transforming data so that it has a **mean of 0** and a **standard deviation of 1**. It allows us to compare values across different datasets or variables measured on different scales.

## Formula

For a data point x:

$$Z = \frac{x - \mu}{\sigma}$$

where:

- x = individual data value
- μ = mean of the dataset
- σ = standard deviation of the dataset

## Standardization Transforms a Dataset

- **Centers the data**: Subtracting the mean shifts the dataset so the average becomes 0.
- **Scales the data**: Dividing by the standard deviation adjusts the spread so the variability is standardized to 1.

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- **Result**: The transformed dataset follows a **standard normal distribution** (mean = 0, SD = 1).

## Importance in Data Analysis

- **Comparability**: Makes variables measured in different units (e.g., height in cm, weight in kg) directly comparable.
- **Probability calculations**: Allows use of standard normal distribution tables to compute probabilities.
- **Machine learning**: Many algorithms (like regression, clustering, neural networks) perform better when features are standardized.
- **Outlier detection**: Z-scores highlight how far a value is from the mean in terms of standard deviations.

  Standardization converts raw values into a common scale, enabling fair comparison and easier statistical analysis.

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**Question no 5:-** What is Kurtosis and their type?

**Answer:-** **Kurtosis:**

Kurtosis is a statistical measure that describes the shape of a distribution's tails compared to a normal distribution. It tells us whether data are heavy-tailed (more extreme values) or light-tailed (fewer extreme values).

Mathematically, kurtosis is based on the fourth central moment of a distribution.

## Types of Kurtosis

1. **Mesokurtic (Normal Kurtosis)**
   - Distribution has kurtosis ≈ 3 (same as the normal distribution).
   - Tails are neither too heavy nor too light.
   - Example: Standard normal distribution.
2. **Leptokurtic (High Kurtosis)**
   - Kurtosis > 3.
   - Distribution has heavy tails and a sharper peak.
   - More prone to outliers.
   - Example: Student's t-distribution with low degrees of freedom.

3. **Platykurtic (Low Kurtosis)**

   i. Kurtosis < 3.
   ii. Distribution has **light tails** and a **flatter peak**.
   iii. Fewer extreme values compared to normal.

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iv. Example: Uniform distribution.

Kurtosis helps us understand the likelihood of extreme values (outliers) in a dataset.

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**Question no 6:-** Explain why the uniform distribution is a good model for the outcome of rolling a fair die.

**Answer:-** A uniform distribution is a good model for the outcome of rolling a fair die because each face (1 through 6) has an equal probability of appearing.

### 🔹 Why Uniform Distribution Fits

- Equal likelihood: In a fair die, no number is favored — each outcome has probability 16.
- Range of outcomes: The possible values are discrete integers from 1 to 6.
- Flat probability distribution: The probability mass function assigns the same value to each outcome, which is exactly what a uniform distribution represents.

### 🔹 Key Characteristics

- Discrete uniform distribution: Since the die produces whole numbers (1–6), this is a discrete case of uniform distribution.
- Shape: If you plot the probabilities, the histogram is flat — all bars are the same height.
- Expectation: The mean is 1+62=3.5, and the variance is (n2−1)12=3512 for n=6.

The uniform distribution is ideal because it mathematically captures the fairness of the die: every outcome is equally likely, no bias, no skew.

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**Question no 7:-** Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5

**Answer:-** Identify parameters

- Number of trials (n) = 15
- Probability of success (p) = 0.5
- We want: P(X≥8)

## Formula in Excel

We can use the BINOM.DIST function. For exact probability of k successes:

Code

**=BINOM.DIST(k, 15, 0.5, FALSE)**

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For cumulative probability up to k:

Code

**=BINOM.DIST(k, 15, 0.5, TRUE)**

## Compute P(X≥8)

This is the complement of P(X≤7):

P(X≥8)=1−P(X≤7)

In Excel:

Code

**=1 - BINOM.DIST(7, 15, 0.5, TRUE)**

The probability of getting **at least 8 successes in 15 trials** with p=0.5 is approximately **0.40 (40%)**.

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**Question no 8:-** How does log transformation help in stabilizing variance and making data more normally distributed?

**Answer:-** **Concept of Log Transformation**

A logarithmic transformation is a mathematical technique applied to data where each value x is replaced by log(x). It's commonly used in statistics and data analysis to handle skewed data and stabilize variance.

## It Helps

1. Stabilizing Variance (Reducing Heteroscedasticity)
   - In many datasets, variance increases with the mean (e.g., income, population sizes).
   - Taking logs compresses large values more than small ones, reducing the spread and making variance more uniform across the dataset.
2. Making Data More Normally Distributed
   - Many real-world datasets (like salaries, reaction times, or biological measures) are positively skewed.

○ Log transformation reduces skewness by pulling in the long right tail, making the distribution closer to symmetric (normal).

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- This is important because many statistical tests (t-tests, regression, ANOVA) assume normality.

## Formula

For a value x>0:

y=log(x)

- Often natural log (ln) is used, but base-10 or base-2 logs can also be applied depending on context.

## Effect on Dataset

- Original data: Skewed, with large outliers dominating.
- After log transformation:
  - Mean shifts downward.
  - Variance becomes more stable.
  - Distribution looks more symmetric and bell-shaped.

## Example

- Suppose incomes in a dataset range from ₹10,000 to ₹1,000,000.
- Raw data: highly skewed, with a few very high earners.
- After log transformation: the extreme values are compressed, making the distribution more balanced and suitable for statistical modeling.

Log transformation is a powerful tool to tame skewed data, stabilize variance, and align datasets with the assumptions of normal-based statistical methods.

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# Distribution